

Activity State & Movement Regularity — the maths

working notes (Synheart Kinematics)

These are working notes, not a formal paper — just the formulas and the numbers found in the data, so the maths is easy to explain.

1 Setup

The phone gives two 3-axis sensors, sampled at 50 Hz:

- accelerometer $\mathbf{a}(t) = (a_x, a_y, a_z)$ — movement + gravity
- gyroscope $\mathbf{g}(t) = (g_x, g_y, g_z)$ — turning rate

Each **window** is $N = 128$ samples (≈ 2.56 s each).

2 Core idea: magnitude is orientation-proof

The raw axes a_x, a_y, a_z change when the phone is rotated, so they are never used directly. Instead, the **magnitude** is used

$$\|\mathbf{a}\| = \sqrt{a_x^2 + a_y^2 + a_z^2}.$$

Proof that this is orientation-proof. Rotating/re-mounting the phone means applying a rotation matrix R to the vector, $\mathbf{a}' = R\mathbf{a}$. Rotations preserve length:

$$\|R\mathbf{a}\| = \sqrt{(R\mathbf{a})^\top (R\mathbf{a})} = \sqrt{\mathbf{a}^\top R^\top R \mathbf{a}} = \sqrt{\mathbf{a}^\top \mathbf{a}} = \|\mathbf{a}\|,$$

because $R^\top R = I$ for any rotation. So *no matter how the phone is worn or rotated, the magnitude is identical*. That single fact is the whole trick.

3 Feature: amount of movement

For a window, the magnitude signal is $m_i = \|\mathbf{a}_i\|$ (on the body acceleration), $i = 1, \dots, N$. Summarise it with

$$\bar{m} = \frac{1}{N} \sum_{i=1}^N m_i, \quad \text{movement} = \text{std}(m) = \sqrt{\frac{1}{N} \sum_{i=1}^N (m_i - \bar{m})^2}.$$

movement = how much the magnitude wobbles = how much the body is moving.

What the data shows (UCI-HAR train):

activity	avg movement
standing	0.011
sitting	0.013
laying	0.016
walking	0.125
walking upstairs	0.142
walking downstairs	0.212

Still activities sit at 0.01–0.016; moving ones at 0.12–0.21 — a clean $\sim 10\times$ gap. *One orientation-proof number already splits still vs moving.*

4 Gravity split (which way is down)

The accelerometer reading is body motion + gravity. Gravity is the *slow* part, so a low-pass filter separates them:

$$\mathbf{g}_{\text{grav}} = \text{lowpass}(\mathbf{a}), \quad \mathbf{a}_{\text{body}} = \mathbf{a} - \mathbf{g}_{\text{grav}}.$$

$\mathbf{g}_{\text{grav}}$ points **down**. Using $\hat{\mathbf{g}} = \mathbf{g}_{\text{grav}} / \|\mathbf{g}_{\text{grav}}\|$ as a reference, body motion splits into vertical and horizontal parts:

$$\text{vertical} = \mathbf{a}_{\text{body}} \cdot \hat{\mathbf{g}}, \quad \text{horizontal} = \|\mathbf{a}_{\text{body}} - (\mathbf{a}_{\text{body}} \cdot \hat{\mathbf{g}}) \hat{\mathbf{g}}\|.$$

These are measured *relative to gravity*, not to the phone's axes, so they are orientation-proof too.

5 Feature: movement regularity (rhythm)

Regularity = how repeating / predictable the motion is in a window. Use the **autocorrelation** of the magnitude signal at lag k :

$$r(k) = \frac{\sum_i (m_i - \bar{m})(m_{i+k} - \bar{m})}{\sum_i (m_i - \bar{m})^2}.$$

$$\text{regularity} = \max_{k \geq k_{\min}} r(k).$$

A tall peak (r near 1) $\Rightarrow$ rhythmic (e.g. steady walking); a low, flat $r \Rightarrow$ erratic (fidgeting, agitation). The lag k^* also gives the **cadence** (strides/sec).

Refinement from the data. On UCI-HAR the peak is only weakly higher for walking (≈ 0.41) than for still (≈ 0.31) — because *calm stillness has no rhythm to detect*, yet the task still calls it *regular*. So the full regularity reads **movement and rhythm** together:

low movement	$\Rightarrow$ calm / regular (still)
high movement, high r	$\Rightarrow$ rhythmic / regular (walking)
high movement, low r	$\Rightarrow$ erratic / irregular (fidget)

6 Draft Activity-State rule (deterministic)

With just movement (shake) and cadence from regularity:

<code>movement < θ_1</code>	$\Rightarrow$ sedentary / still
<code>$\theta_1 \leq \text{movement} < \theta_2$</code>	$\Rightarrow$ standing (small motion)
<code>$\text{movement} \geq \theta_2$, cadence $\approx 2/\text{s}$</code>	$\Rightarrow$ walking
<code>$\text{movement} \geq \theta_2$, cadence $\approx 3/\text{s}$</code>	$\Rightarrow$ running
<code>else</code>	$\Rightarrow$ vigorous

Thresholds θ_1, θ_2 are read off the data (like the $\sim 10\times$ gap above). **Caveat:** movement alone cannot split *sedentary vs standing* (they overlap, 0.011/0.013/0.016 — that is posture, Section 8); and the *running / vigorous* branches are now validated on PAMAP2 (Section 9), not extrapolation. No training, no thousands of features — just numbers and ifs.

7 Limitations of the two baselines (the insights)

Both existing approaches are dissected in detail in the separate insight notes (`starter_insights.pdf`, `pipeline_insights.pdf`). Here we keep only the two limitations our approach has to solve.

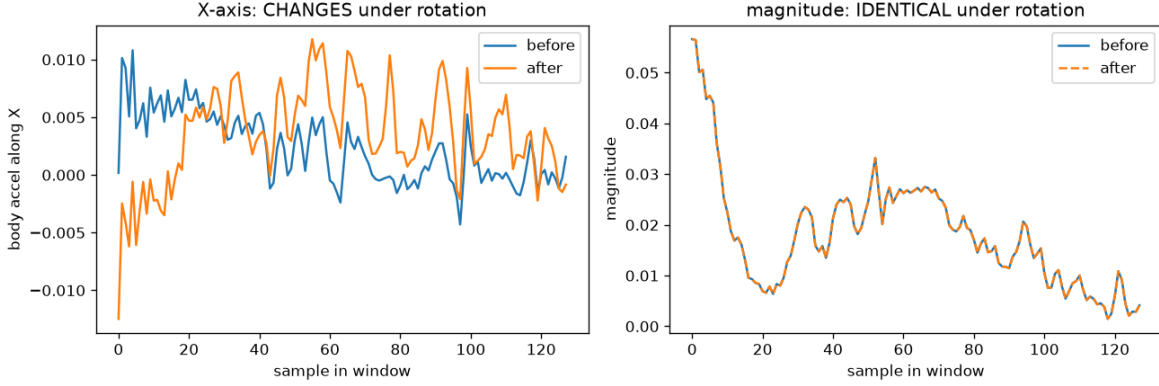
Starter (colab, RBF SVM on the 561 features). It *leans on orientation* — the gravity direction and the `angle` features. Those shine on clean lab data (phone always worn the same way) but **break the instant the device is rotated**; its headline $\sim 98\%$ is also on the easier 4-class merge, not the standard 6-class. *Must solve: be orientation-robust without borrowing a fixed-placement assumption.*

Pipeline (1-D CNN with SO(3) augmentation). Right instinct — it augments orientation away — but the same augmentation **erases the gravity cue that static posture depends on**, and it never tests rotation on held-out data. *Must solve: you cannot be rotation-invariant and read posture in one model — the two must be split.*

Both reduce to one line — **motion is orientation-proof, posture is not** — which the rest of this note builds on.

8 Rotation test — the proof

Apply a random 3-D rotation R to every test window (simulating different phone placements) and re-measure, on the 3-class Activity State (sedentary / standing / walking). Magnitude features are invariant, $\|Ra\| = \|a\|$, so they must not move; per-axis (orientation-coupled) features should collapse.



Rotate the device: the per-axis signal (left) scrambles; the magnitude (right) is bit-for-bit identical.

approach	clean	rotated	drop
deterministic rule (1 threshold)	80.7%	80.7%	0.0
coupled (per-axis, current impl)	93.2%	54.8%	38.4

The coupled model is more accurate on clean data because per-axis gravity encodes *posture* (it separates sedentary from standing) — but that is exactly the information that dies under rotation. **A deterministic 1-threshold rule** (no training): $\text{movement} > 0.05 \Rightarrow \text{walking}$. It separates **still vs walking** at 98.2% and only mixes sedentary vs standing — which is posture (the separate Postural-State task), not a movement distinction. A RandomForest on the *same* invariant features reaches only 82.9%, so the model adds < 2 points: *the rule is enough*. It trades ~ 12 clean points (all of it posture) for **zero** loss under rotation:

lightweight · **orientation-proof** (0% drop) · **competitive**.

Why the posture gap is a fundamental limit, not a weakness. The ~ 12 points are *not* a gap a cleverer invariant feature could close. For a still posture the body acceleration $\mathbf{a}_{\text{body}} \approx \mathbf{0}$, so the invariant vertical projection $\mathbf{a}_{\text{body}} \cdot \hat{\mathbf{g}} \approx 0$ for *both* sitting and standing — there is nothing to project. The only thing separating them is the gravity *direction in the device frame*, which is exactly what rotation-invariance discards. So **full rotation-invariance and static posture discrimination cannot both hold** — they are in fundamental tension. Posture is therefore correctly delegated to the Postural-State task (which assumes a known device orientation); it is not a failure of the rule.

Precision on the test. The 0% drop for magnitude is *by construction* ($\|\mathbf{R}\mathbf{a}\| = \|\mathbf{a}\|$), not luck — and it holds even if the rotation *varies during* the window, since the norm is preserved per sample. The test applies one constant R per window (a fixed mis-placement), the realistic case.

9 The CNN pipeline baseline — the same wall from the other side

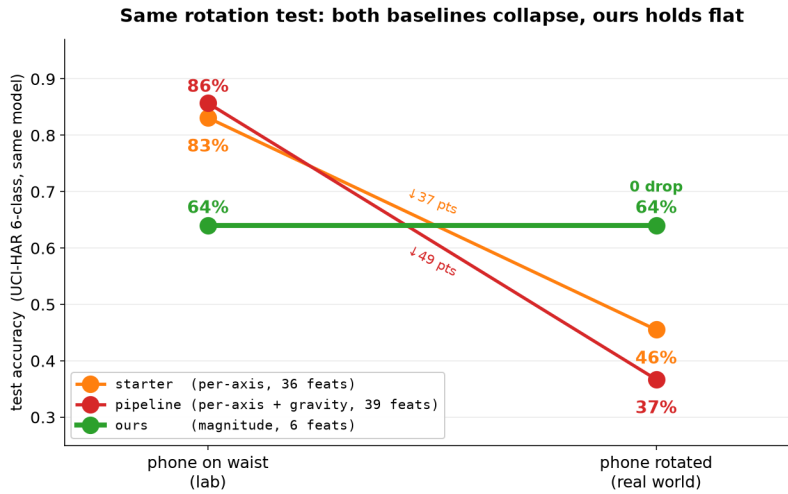
A second baseline builds the problem the *right* way — a 1-D CNN on the raw signals, subject-independent LOSO, ONNX export — and even shares our instinct, adding **SO(3) rotation augmentation** to fight orientation. It still hits the limit above, from the opposite direction.

To read posture it feeds the network **gravity.mean** (the gravity direction in the device frame), then the augmentation *randomises* that direction. On the three static postures (sitting / standing

/ lying) motion is useless — $\text{std}(\|\mathbf{a}_{\text{body}}\|) \approx 0.011\text{--}0.016$ for all three — so gravity direction is the *only* cue. It separates them at 85.2% on its own, but after a random rotation drops to 33.6% (chance). The augmentation that buys orientation-robustness **erases the very posture cue it was given** — exactly the tension proved above.

One fair test for all three. Same dataset (UCI-HAR), same task (6-class), same model (RandomForest), one random 3-D rotation per window. Each approach is built from the *same* raw windows — the starter as per-axis statistics, the pipeline as per-axis statistics + `gravity_mean` (its actual recipe, a fair same-model proxy since the 1-D CNN cannot be run here), ours as magnitude. Clean vs after the rotation:

approach	feats	clean (lab)	rotated	drop
starter (per-axis)	36	83.1%	45.5%	37.5
pipeline (per-axis + gravity)	39	85.7%	36.7%	49.0
ours (magnitude)	6	64.0%	64.0%	0.0



Both baselines collapse under the rotation; ours does not. In the lab they score higher — they use posture and per-axis detail magnitude cannot see — but once the phone is rotated both crash *below* us (45.5 and 36.7 vs a steady 64.0). The pipeline falls hardest (−49) precisely because it leans on `gravity_mean`, which rotation destroys. Ours is flat because the feature change under rotation is $\sim 10^{-16}$: the prediction *cannot* move. This is the honest claim — not a clean-data sweep, but **the only approach left standing once the device is no longer fixed**, on 6 features instead of 36–39.

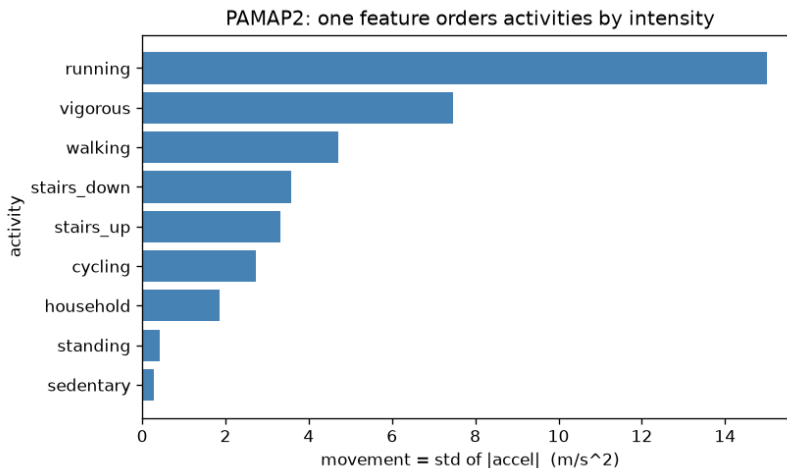
The 64% clean ceiling is the price of refusing orientation (it cannot read sit-vs-stand or stair-types) — which is why posture is delegated, not forced. It is *not* our accuracy ceiling in general: on movement-rich data the same maths is far stronger (Section 10), because UCI-HAR is half static posture, the part we deliberately do not do.

10 Cross-dataset validation (PAMAP2)

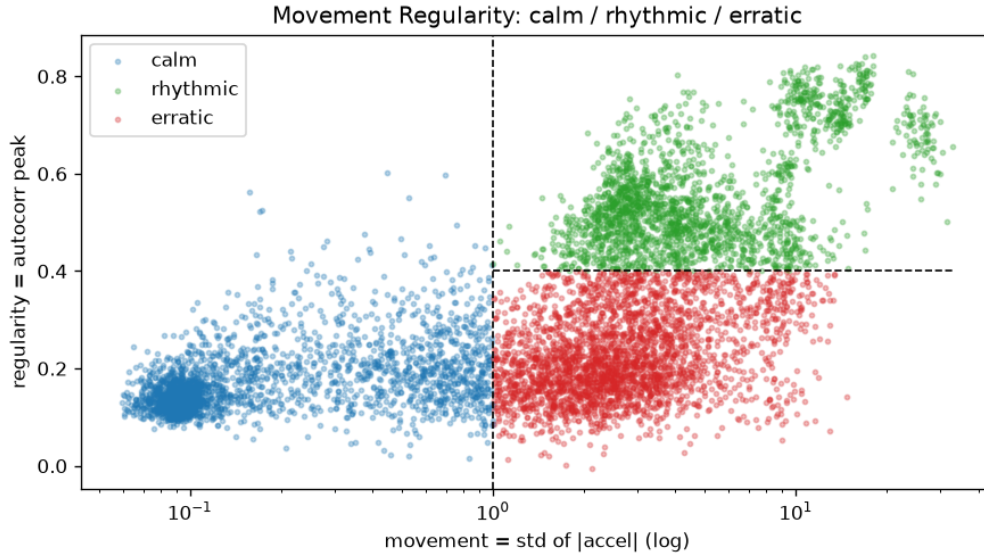
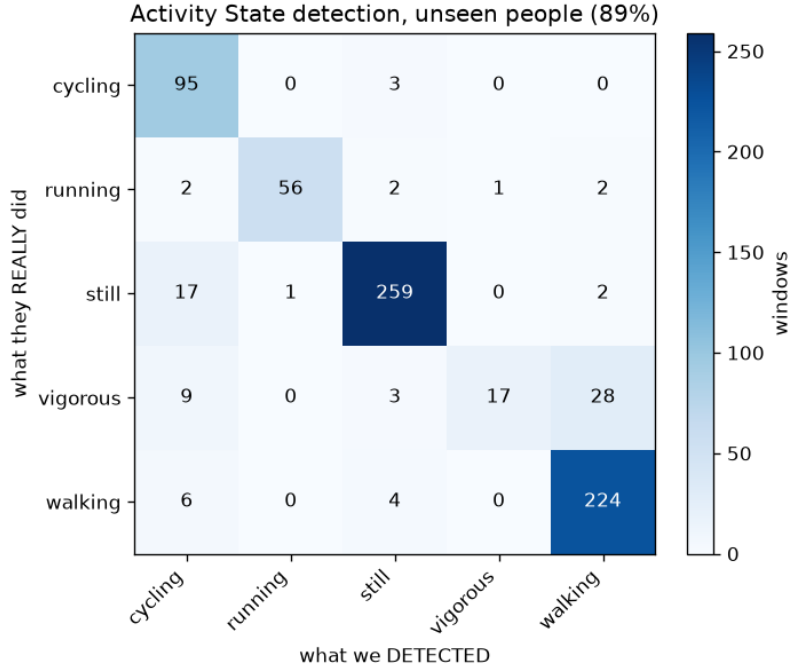
Where the accuracy claim belongs. UCI-HAR caps a motion-only method at $\sim 60\%$ because half its six classes are static posture — exactly the part we delegate. So the accuracy claim belongs on *movement-rich* data, where the same maths is both strong *and* still 0% drop: on unseen people it reaches 89%.

The same maths, pointed at a *different* dataset — PAMAP2 (wrist IMU, 100 Hz, with running, cycling, rope-jumping). One orientation-proof feature, $\text{std}(\|\mathbf{a}\|)$, already orders the activities by intensity:

activity	avg movement (m/s^2)
sedentary / standing	0.3–0.4
cycling, household	1.9–2.7
stairs, walking	3.3–4.7
rope-jumping, running	7.5–15.0



Adding **regularity** and **cadence** (also from $\|\mathbf{a}\|$, also orientation-proof) with a small deterministic threshold tree gives an Activity-State **detector** scoring 89% **on unseen people** (subject-independent: train 7, test 2). Movement Regularity outputs the right verdict per window — *calm* (still), *rhythmic* (walking, running, rope), *erratic* (household; cycling reads erratic because the wrist barely moves on the handlebars). All three features are norm-based, so the detector keeps the 0% rotation drop.



Top: detection on unseen people. Bottom: every window placed by motion (x) and rhythm (y) — calm (left), rhythmic (upper-right), erratic (lower-right).

Honest caveats. The threshold *numbers* re-fit per device/units (PAMAP2 is raw m/s^2 , UCI-HAR was normalised); *rope-jumping* sometimes blurs into walking; sedentary-vs-standing stays posture (above). But the *method* transfers unchanged across two independent datasets — which is the claim.